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3310 - Project

Ranking Teams Using Markov Chains

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Abstract:

Ranking the best sport teams is a very subjective process. In the Big 12 Men's Basketball Conference, they determine their standing strictly based on win-loss record. This is a common and prudent way to rank teams; however, it ignores context and margin of victory. This project applies the Markov chain to create a more nuanced ranking of the 16 teams in the Big 12 using the results of the 2025-2026 conference play.

To execute this, we create an irreducible transition matrix made by using the point differential and other statistics from each game. For each matchup, a dominance value is calculated by taking the differential of a specific statistical category. The team that outperforms their opponent in that metric is rewarded the dominance value, and the other team receives zero. These values combine to form a transition matrix that is normalized into a column stochastic matrix where every element is a probability and each column sums to one. From this stochastic matrix, we determine each team's ranking from their eigenvector corresponding to eigenvalue $\lambda = 1$. The results are a more refined ranking for the best teams in the conference.

Attribution:

Eshaan Murali did the abstract, introduction, and wrote the code. Henry Matar worked on the Mathematical Formulation and Examples & Numerical Results section. Ashlyn Emery worked on the Discussion and Conclusion.

Introduction:

In this project, we utilize the methods established in the paper "A Markov Method for Ranking College Football Conferences" by R. Bruce Mattingly and Amber J. Murphy to merge concepts of linear algebra with college basketball standings. Using the techniques in the paper, we wanted to develop our own Markov Chain model that would rank the 16 teams in the Men's Big 12 Conference. To accomplish this, we had to understand how a Markov Process works and how we can apply it to ranking basketball teams that play against each other a different number of times. This paper helped us understand the concepts of a transition matrix which helped us with incorporating our own transition matrix for the 16 teams in the Big 12.

A Markov Chain is a stochastic process where the probability of moving to a future state depends only on the current state. To start this process, we compiled results of every conference game and treated each game as a link between the two teams. When one team wins, it creates a chain between the two where the credit of the chain is the point differential of the result. To transform these point differentials into probabilities, we normalized the data by dividing each column by the total number of points that team lost. This ensured that every column summed to one, making our matrix a column stochastic matrix. All these results put together resulted in a 16x16 transition matrix from which each entry was a probability linked between teams. Once we have our transition matrix, we find its steady state vector. The steady state vector is the eigenvector that corresponds to eigenvalue $\lambda = 1$. This steady state vector represents the probability distribution where the team with the highest values are the best teams in the conference.

Mathematical Formulation:

By modeling the Big 12 season as a Markov chain, where each team is a state and the results establish transition probabilities between states. The chain produces a rating that models the strength of every team considering both margin of victory (point differential) and quality of opponents.

Raw Transition Matrix Construction:

There are 16 teams in the Big 12, let each team be indexed i=1,2...16. From the results of in conference games, we will construct a 16x16 matrix, A, whose entries record the margin in which team i defeated team j (e.g., BYU beats Colorado by 4 points, so 4 goes into the i,j entry).

$$a_{i,j} = \begin{cases} (stat\ i\ scored\ against\ j) - (stat\ j\ scored\ against\ i), & if\ i > j \\ 0, & else \end{cases}$$

It's important to note that a team cannot play itself; therefore, $a_{i,i} = 0$.

Normalizing the Transition Matrix:

The raw matrix is converted into the transition matrix by dividing each entry by its column sum. Let the normal matrix be P with its entries being $P_{i,j}$.

$$p_{i,j} = \frac{a_{i,j}}{\sum_{k=1}^{16} a_{k,j}}$$

This operation will divide the raw entry by the sum of its column, returning a value between 0 and 1 and the column sum of the transition matrix $\sum_{i=1}^{16} p_{i,j} = 1$.

Uncovering the Ranking Vector **V**:

Using the transition matrix, we need to find a vector, **v,** that assigns a strength rating to each team. We need an eigenvector with an eigenvalue of λ=1, so the vector doesn't change when multiplied by matrix P. This vector will converge regardless of its initial value.

Computing Dominant Eigenvector with the Power Method:

V cannot be interpreted from matrix P, so the dominant eigenvector must be isolated using the power method.

$$v_{n+1} = \frac{Pv_n}{||Pv_n||}$$

The result is the dominant eigenvector which is our rating vector. We start with an initial assumption that all teams are equally strong and iterate until all entries converge, meaning start with an initial matrix of ones then normalize it. Ex:

$$v_0 = \frac{1}{16}\begin{bmatrix} 1 \\ 1 \\ . \\ . \\ 1 \end{bmatrix}$$

Convergence occurs when $||v_{n+1} - v_n|| < \varepsilon$ (ε is a chosen tolerance).

Interpreting the Vector:

When v converges, all teams are ranked in descending order. Teams with the largest entries accumulated more weight through the Markov chain, indicating their strength, reflecting a potentially higher quality rating than a simple win-loss record.

Examples and Numerical Results:

We take the results from every conference game that occurred during the 2025-2026 Big 12 season and populate a Transition Matrix and Steady-State vector for each statistic (Points, Rebounds, Assists, Steals, Blocks, and Turnovers). Traditional basketball emphasizes these statistics as key measures of a team's performance. Once we derived the matrices and the vectors, we created our final ranking by taking the average ranking from all 6 statistics.

Team Name Abbreviations							
Arizona	**Arizona State**	**Baylor**	**Brigham Young**	**Cincinnati**	**Colorado**	**Houston**	**Iowa State**
UA	ASU	BAY	BYU	CIN	CU	HOU	ISU
Kansas	**Kansas State**	**Oklahoma State**	**Texas Christian**	**Texas Tech**	**Central Florida**	**Utah**	**West Virginia**
KU	KSU	OKST	TCU	TTU	UCF	UU	WVU

Fig. 1 Team Abbreviations

Points Differential

Point differential is the strongest and clearest measure of dominance in a game. A large margin lets one know which team was stronger that night with little uncertainty. The transition matrix presents how much of a scoring advantage one team had to another. Higher values mean a more decisive victory, while dashes represent a loss and therefore have no value to contribute.

BIG 12 MARKOV TRANSITION MATRIX -- POINTS DIFFERENTIAL

	UA	ASU	BAY	BYU	CIN	CU	HOU	ISU	KU	KSU	OKST	TCU	TTU	UCF	UU	WVU
Arizona	--	0.103	0.049	0.101	0.260	0.055	0.375	0.243	0.217	0.106	0.209	0.169	--	0.201	0.098	0.246
Arizona State	--	--	0.042	--	0.140	--	--	--	0.094	0.013	0.051	--	0.089	--	0.108	--
Baylor	--	0.026	--	--	--	0.104	--	--	--	--	0.085	--	--	0.007	0.134	0.070
Brigham Young	--	0.144	0.035	--	--	0.022	--	0.135	--	0.102	--	0.078	0.107	--	0.093	0.141
Cincinnati	--	--	0.070	0.222	--	0.049	--	0.122	0.151	0.123	0.130	--	--	0.139	0.057	--
Colorado	--	0.072	--	--	--	--	--	--	--	0.038	0.079	0.338	--	--	0.134	--
Houston	--	0.155	0.246	0.182	0.290	0.219	--	--	0.208	0.060	0.040	0.117	0.071	0.167	0.072	0.204
Iowa State	--	0.361	0.092	--	--	0.164	0.094	--	0.170	0.145	0.141	--	0.393	0.208	0.082	0.148
Kansas	0.571	--	0.127	0.081	--	0.033	0.406	0.284	--	0.183	0.068	0.117	0.054	--	0.062	--
Kansas State	--	--	0.113	--	--	--	--	--	--	--	--	--	--	--	0.015	0.014
Oklahoma State	--	--	--	0.071	--	0.049	--	--	--	0.004	--	--	--	0.125	0.062	0.049
TCU	--	0.062	0.092	--	0.100	--	--	0.095	--	0.047	0.073	--	0.143	--	--	0.042
Texas Tech	0.429	--	0.134	0.131	0.120	0.197	0.125	0.122	--	0.119	0.124	--	--	--	0.072	0.049
UCF	--	0.015	--	0.131	0.020	0.049	--	--	0.057	0.038	--	0.143	0.143	--	0.010	--
Utah	--	--	--	--	--	--	--	--	--	--	--	0.039	--	--	--	0.035
West Virginia	--	0.062	--	0.081	0.070	0.060	--	--	0.104	0.021	--	--	--	0.153	--	--

Fig. 2 Point Differential Transition Matrix

Then applying the power method to this transition matrix results in the steady-state vector below. Each score $\pi(i)$ is the probability mass the Markov chain delegates to a team. A higher score is evidence of a program that won consistently, by large margins, and against strong opponents.

POINTS DIFFERENTIAL – Steady–State Vector

Rk	Team	pi(i)
1	Kansas	0.1875
2	Arizona	0.1627
3	Iowa State	0.1220
4	Texas Tech	0.1189
5	Houston	0.1049
6	Cincinnati	0.0619
7	Brigham Young	0.0442
8	UCF	0.0419
9	TCU	0.0402
10	Arizona State	0.0382
11	West Virginia	0.0372
12	Colorado	0.0177
13	Oklahoma State	0.0113
14	Baylor	0.0071
15	Utah	0.0029
16	Kansas State	0.0014

Fig. 3 Point Differential Steady-State Vector

Point differential ranking will most closely mirror Big 12 standings as winning teams will place higher in this traditional statistic category. One can see that Kansas ranks first (0.1875) with a noticeable gap between the top 5 and the rest of the conference. Kansas State's last place outcome only accumulated (0.0014). KU accumulated 134x more probability mass than KSU and thus has 134x more strength relative to KSU. This reflects that these bottom programs lost frequently to both strong and weak opponents by significant margins, as the Markov chain punishes greater losses more heavily than close games.

Rebounds Differential

Rebounding is both an offensive and defensive stat, measuring physical presence and possession of the ball. Rebounds create second chances on offense and prevent them on defense. A vital advantage that can compound throughout the game. The transition matrix represents teams that won the battle on the glass repeatedly against an opposing program.

BIG 12 MARKOV TRANSITION MATRIX -- REBOUNDS DIFFERENTIAL

	UA	ASU	BAY	BYU	CIN	CU	HOU	ISU	KU	KSU	OKST	TCU	TTU	UCF	UU	WVU
Arizona	--	0.240	0.137	0.237	0.294	0.031	0.024	0.218	0.558	0.146	0.220	0.140	--	0.226	0.129	0.227
Arizona State	--	--	0.137	--	--	--	--	--	--	0.073	0.132	--	--	--	--	--
Baylor	--	0.017	--	--	0.039	0.054	0.220	0.036	--	0.067	0.011	0.211	0.062	0.016	0.108	0.030
Brigham Young	--	0.058	0.059	--	--	0.039	--	0.182	--	0.129	0.044	0.316	0.167	--	0.226	0.045
Cincinnati	--	0.041	--	0.132	--	0.008	0.024	0.145	0.173	0.039	0.099	0.140	--	0.129	0.032	0.061
Colorado	--	0.165	--	--	--	--	--	--	--	0.045	--	0.018	--	--	0.151	--
Houston	0.667	0.058	0.235	0.211	0.078	0.101	--	0.055	0.135	0.011	--	0.018	--	0.226	0.065	0.167
Iowa State	--	0.091	0.157	--	--	0.124	--	--	--	0.096	0.209	--	0.125	0.129	--	0.136
Kansas	--	0.074	0.059	--	--	0.093	0.171	0.109	--	0.135	0.011	0.035	0.021	--	0.054	--
Kansas State	--	--	--	--	--	--	--	--	--	--	0.011	--	--	--	--	--
Oklahoma State	--	--	--	--	--	0.109	0.073	--	--	--	--	--	--	0.097	0.097	0.045
TCU	--	0.050	0.216	--	--	--	--	0.091	0.077	0.101	0.165	--	0.292	--	--	0.076
Texas Tech	0.333	0.025	--	--	0.275	0.225	0.488	0.145	--	0.067	0.099	--	--	--	0.140	--
UCF	--	0.099	--	0.184	0.020	0.078	--	--	--	0.006	--	0.018	0.250	--	--	0.091
Utah	--	0.083	--	--	0.216	--	--	0.018	--	0.022	--	0.105	--	0.065	--	0.121
West Virginia	--	--	--	0.237	0.078	0.140	--	--	0.058	0.062	--	--	0.083	0.113	--	--

Fig. 4 Rebound Differential Transition Matrix

The steady-state vector is vastly different than the point differential above, telling a completely different narrative than the scoring outcomes of games.

REBOUNDS DIFFERENTIAL — Steady-State Vect

Rk	Team	pi(i)
1	Houston	0.1596
2	Texas Tech	0.1530
3	Arizona	0.1316
4	TCU	0.0762
5	Brigham Young	0.0743
6	Baylor	0.0713
7	UCF	0.0604
8	Cincinnati	0.0544
9	Iowa State	0.0514
10	Kansas	0.0466
11	West Virginia	0.0453
12	Utah	0.0311
13	Oklahoma State	0.0235
14	Arizona State	0.0129
15	Colorado	0.0082
16	Kansas State	0.0003

Fig. 5 Rebound Steady-State Vector

Kansas, which was 1st in fig. 3 has now dropped to the 10th position. Whereas Houston jumped 4 spots to reach the first rank. The top three teams have accumulated noticeably higher probability mass than the other 13, reflecting that Houston, Texas Tech, and Arizona were significantly the most dominant rebounders of the conference. Higher ranked teams suggest a more physically supreme program and success on both the offensive and defensive sides of the court.

Assists Differential

Assists differential measures efficacy and ball movement, creating high quality shots and teamwork. A more efficient team creates scoring opportunities at an elevated rate, which will illustrate some parallels with the point differential analysis.

BIG 12 MARKOV TRANSITION MATRIX -- ASSISTS DIFFERENTIAL

	UA	ASU	BAY	BYU	CIN	CU	HOU	ISU	KU	KSU	OKST	TCU	TTU	UCF	UU	WVU
Arizona	--	0.073	0.086	0.102	0.333	0.151	--	0.409	0.294	0.037	0.083	0.133	0.250	--	0.077	0.150
Arizona State	--	--	--	--	--	0.094	--	--	--	--	--	--	--	--	0.015	--
Baylor	--	0.109	--	0.045	--	0.113	0.206	--	--	--	0.024	0.067	--	--	0.108	0.033
Brigham Young	0.300	0.018	--	--	--	--	--	--	--	0.061	--	--	--	--	0.031	0.033
Cincinnati	--	0.036	0.057	0.182	--	0.057	0.029	0.182	0.147	0.122	0.107	--	--	0.227	0.108	0.167
Colorado	--	0.036	--	0.023	--	--	--	--	--	0.037	0.155	0.267	--	--	0.108	0.017
Houston	0.600	0.109	0.143	0.080	0.143	0.113	--	--	0.088	0.061	0.071	--	--	0.182	--	0.117
Iowa State	--	0.309	0.229	--	--	0.113	0.029	--	0.147	0.183	0.083	0.117	0.750	0.205	0.031	0.217
Kansas	--	0.036	0.171	0.045	--	--	0.118	0.273	--	0.256	0.083	--	--	0.023	0.123	--
Kansas State	--	--	0.171	--	--	--	--	--	--	--	0.095	0.133	--	0.159	0.108	--
Oklahoma State	--	0.073	--	0.068	--	--	--	--	--	--	--	0.067	--	--	0.077	0.033
TCU	--	0.018	0.057	0.057	0.095	--	0.088	--	0.206	--	0.071	--	--	--	--	0.017
Texas Tech	--	0.018	--	0.159	0.286	0.340	0.382	0.136	0.088	0.134	0.131	0.067	--	--	0.215	0.117
UCF	0.100	0.055	0.086	0.159	0.048	0.019	--	--	--	--	0.095	0.133	--	--	--	0.050
Utah	--	0.073	--	--	--	--	0.147	--	--	--	--	0.017	--	0.068	--	0.050
West Virginia	--	0.036	--	0.080	0.095	--	--	--	0.029	0.110	--	--	--	0.136	--	--

Fig. 6 Assists Differential Transition Matrix

The assists differential steady state vector not only rewards scoring, but ball movement, teamwork, and intelligent decision making.

ASSISTS DIFFERENTIAL – Steady-State Vector

Rk	Team	pi(i)
1	Arizona	0.1637
2	Iowa State	0.1477
3	Houston	0.1391
4	Texas Tech	0.1302
5	Kansas	0.0763
6	Cincinnati	0.0738
7	Brigham Young	0.0520
8	TCU	0.0413
9	Baylor	0.0396
10	UCF	0.0394
11	Utah	0.0250
12	Kansas State	0.0221
13	West Virginia	0.0213
14	Colorado	0.0175
15	Oklahoma State	0.0091
16	Arizona State	0.0020

Fig. 7 Assists Steady-State Vector

Arizona leads the conference in the assists category with a mass of 0.1637. Having another piece of the offensive puzzle is truly meaningful. One can see Arizona is ranked 1st in fig, 7 and 2nd in fig. 3, informing that their offense is built on ball movement and teamwork rather than individual ability. Whereas Kansas falls from 1st to 5th suggesting that their scoring power relies more on individual talent.

Steals Differential

Steals are a very interesting stat as they are both rare and decide clutch moments by forcing turnovers. Teams that make more steals get more opportunities to score while taking that away from their opponent. As a low frequency event in basketball this is a more unpredictable category.

BIG 12 MARKOV TRANSITION MATRIX -- STEALS DIFFERENTIAL

	UA	ASU	BAY	BYU	CIN	CU	HOU	ISU	KU	KSU	OKST	TCU	TTU	UCF	UU	WVU
Arizona	--	0.050	--	0.176	0.138	--	0.583	0.045	0.026	0.095	0.125	0.130	0.026	0.029	0.167	0.267
Arizona State	0.400	--	0.056	--	--	0.095	--	--	--	--	0.042	--	0.132	--	0.028	--
Baylor	--	--	--	--	--	--	--	0.182	--	--	--	0.087	0.079	--	0.194	--
Brigham Young	--	0.025	0.194	--	--	--	--	0.045	--	0.095	--	0.087	0.026	0.029	0.028	0.100
Cincinnati	--	0.025	0.056	0.235	--	0.190	--	0.182	0.079	0.143	--	--	0.053	0.235	0.028	0.033
Colorado	--	0.050	0.056	0.029	--	--	--	--	0.132	--	0.125	0.130	0.026	0.147	0.056	0.033
Houston	0.400	0.100	0.250	0.206	0.483	0.238	--	--	0.105	--	0.083	--	0.132	0.147	0.083	0.167
Iowa State	--	0.400	0.083	--	--	--	0.250	--	0.053	--	0.208	--	0.237	0.353	0.111	--
Kansas	--	0.100	0.167	0.059	--	--	--	0.045	--	--	--	0.217	--	--	--	0.167
Kansas State	--	0.025	0.056	0.088	--	0.048	0.167	0.045	0.158	--	--	0.043	0.053	--	--	0.067
Oklahoma State	--	--	0.028	0.147	--	0.048	--	0.136	0.079	0.143	--	0.304	0.026	--	--	--
TCU	--	0.175	0.028	--	0.172	--	--	0.182	--	0.238	0.125	--	0.079	0.029	0.111	0.133
Texas Tech	--	--	--	--	--	0.190	--	--	0.237	--	--	--	--	--	--	0.033
UCF	0.200	0.025	0.028	--	0.138	--	--	--	0.026	0.143	0.208	--	0.079	--	0.167	--
Utah	--	--	--	0.059	0.034	0.190	--	--	0.105	0.048	0.042	--	0.053	--	--	--
West Virginia	--	0.025	--	--	0.034	--	--	0.136	--	0.095	0.042	--	--	0.029	0.028	--

Fig. 8 Steals Differential Transition Matrix

STEALS DIFFERENTIAL – Steady-State Vector

Rk	Team	pi(i)
1	Houston	0.1492
2	Arizona	0.1462
3	Iowa State	0.1114
4	TCU	0.0734
5	Arizona State	0.0694
6	Cincinnati	0.0670
7	UCF	0.0660
8	Oklahoma State	0.0556
9	Kansas State	0.0504
10	Colorado	0.0403
11	Kansas	0.0399
12	Baylor	0.0323
13	Brigham Young	0.0302
14	West Virginia	0.0289
15	Utah	0.0216
16	Texas Tech	0.0181

Fig. 9 Steals Steady-State Vector

The top 3 teams dominate this stat, while all other teams are far behind, portraying they experience fewer clutch moments and turnovers. A particularly interesting shift is Texas Tech dropping to dead last despite being top 4 in every category so far. This depicts that TTU generates fewer turnovers but dominates the sport through other defensive measures and a strong offense.

Blocks Differential

Blocking is another low frequency event but creates divisive moments. An expresses strong interior defense. Teams that out-block their opponents are better at preventing offenses from attacking in the paint and forcing lower percentage shots on the perimeter. Blocks also indicate a team has presence and size.

BIG 12 MARKOV TRANSITION MATRIX -- BLOCKS DIFFERENTIAL

	UA	ASU	BAY	BYU	CIN	CU	HOU	ISU	KU	KSU	OKST	TCU	TTU	UCF	UU	WVU
Arizona	--	--	--	0.375	0.091	0.067	--	0.136	--	0.089	0.023	--	--	0.179	0.100	0.111
Arizona State	0.176	--	0.045	--	0.091	0.200	--	0.136	0.375	0.156	0.114	0.077	--	0.107	0.120	0.167
Baylor	0.059	--	--	--	--	0.133	--	0.114	--	0.067	0.068	0.115	0.125	--	--	--
Brigham Young	0.118	0.529	--	--	--	0.100	0.250	0.023	0.125	0.133	0.045	--	0.125	0.107	0.080	--
Cincinnati	--	--	0.091	0.500	--	0.067	--	0.045	--	0.044	--	0.154	--	0.143	0.220	0.167
Colorado	--	0.059	--	--	--	--	--	0.023	--	0.111	0.023	--	0.125	--	0.100	0.056
Houston	0.118	0.353	0.182	0.125	0.273	0.033	--	0.068	0.250	0.044	0.091	0.038	0.125	0.036	0.100	0.028
Iowa State	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	0.111
Kansas	0.118	--	0.182	--	0.091	0.133	0.375	0.114	--	0.156	0.159	0.038	0.500	0.179	0.100	0.056
Kansas State	--	--	--	--	--	--	--	0.068	--	--	0.045	0.192	--	--	--	0.083
Oklahoma State	--	--	--	--	--	0.033	--	0.091	--	--	--	--	--	0.036	--	--
TCU	0.235	--	0.091	--	--	--	--	0.023	0.250	--	0.273	--	--	--	0.020	--
Texas Tech	0.176	--	--	--	0.455	0.200	0.375	0.045	--	0.111	0.023	0.154	--	0.071	0.160	0.056
UCF	--	--	0.273	--	--	0.033	--	0.068	--	0.044	--	--	--	--	--	0.083
Utah	--	0.059	0.136	--	--	--	--	0.045	--	0.044	0.114	--	--	--	--	0.083
West Virginia	--	--	--	--	--	--	--	--	--	--	0.023	0.231	--	0.143	--	--

Fig. 10 Blocks Differential Transition Matrix

BLOCKS DIFFERENTIAL – Steady-State Vector

Rk	Team	pi(i)
1	Kansas	0.1552
2	Houston	0.1510
3	Brigham Young	0.1392
4	Texas Tech	0.1290
5	Arizona State	0.0971
6	Cincinnati	0.0906
7	Arizona	0.0684
8	TCU	0.0584
9	Baylor	0.0314
10	Colorado	0.0254
11	West Virginia	0.0151
12	Kansas State	0.0127
13	Utah	0.0121
14	UCF	0.0114
15	Iowa State	0.0017
16	Oklahoma State	0.0014

Fig. 11 Blocks Steady-State Vector

Kansas is back to the top (0.1552), alongside Houston (0.1510) and BYU (0.1392). This is BYU's first top three finish across all categories. BYU likely has a strong interior defense but possesses weaknesses on other parts of the court. Iowa State drops to 15th its lowest ranking across all categories, suggesting a lack of size, but retaining strengths elsewhere. Also, this is another dead last finish for Oklahoma State, where they finished last in points too.

Turnover Differential

Turnover differential is inverted in our analysis. A team wins the differential when they end a game with fewer turnovers. Therefore, a winning team is committing fewer errors and protecting the ball better.

BIG 12 MARKOV TRANSITION MATRIX -- TURNOVERS DIFFERENTIAL

	UA	ASU	BAY	BYU	CIN	CU	HOU	ISU	KU	KSU	OKST	TCU	TTU	UCF	UU	WVU
Arizona	--	--	--	0.018	0.148	--	0.778	--	--	0.053	0.032	0.154	0.019	0.070	0.056	0.140
Arizona State	0.182	--	0.038	0.036	0.056	0.033	--	--	0.042	--	0.194	--	0.130	--	0.167	--
Baylor	0.091	0.043	--	--	--	--	--	0.222	--	--	--	--	0.037	--	0.241	--
Brigham Young	0.091	--	0.151	--	--	--	--	--	--	0.079	--	--	0.093	0.163	--	0.209
Cincinnati	--	--	0.038	0.164	--	0.200	--	0.278	0.021	0.079	0.194	--	0.148	0.140	0.074	0.163
Colorado	--	0.021	--	0.164	--	--	--	--	0.167	--	0.032	0.154	0.037	0.093	0.056	0.023
Houston	--	0.149	0.377	0.091	0.370	0.333	--	--	0.167	0.053	0.097	0.154	0.185	0.140	0.111	0.209
Iowa State	0.364	0.574	0.038	0.018	--	--	0.222	--	0.146	0.053	0.194	--	0.148	0.279	0.185	0.023
Kansas	0.273	--	0.094	0.018	--	--	--	0.222	--	--	--	0.308	--	--	--	0.047
Kansas State	--	0.064	0.019	0.109	--	0.100	--	--	0.167	--	--	0.077	--	--	--	0.070
Oklahoma State	--	--	0.038	0.145	--	0.033	--	0.167	0.062	0.237	--	0.077	--	--	--	--
TCU	--	0.106	0.170	--	0.222	--	--	0.111	--	0.289	0.065	--	0.019	--	0.037	0.116
Texas Tech	--	--	--	0.036	--	0.133	--	--	0.125	--	0.032	--	--	--	0.019	--
UCF	--	0.021	0.019	--	0.204	--	--	--	0.104	--	0.161	0.077	0.130	--	--	--
Utah	--	--	--	0.109	--	0.167	--	--	--	0.053	--	--	--	--	--	--
West Virginia	--	0.021	0.019	0.091	--	--	--	--	--	0.105	--	--	0.056	0.116	0.056	--

Fig. 12 Turnovers Differential Transition Matrix

The turnover differential steady-state vector awards discipline, intelligent playmaking, and reflects both offensive and defensive ability.

TURNOVERS DIFFERENTIAL – Steady-State Vector

Rk	Team	pi(i)
1	Iowa State	0.1495
2	Arizona	0.1280
3	Houston	0.1220
4	Kansas	0.0944
5	Cincinnati	0.0847
6	TCU	0.0653
7	Oklahoma State	0.0521
8	Arizona State	0.0512
9	Baylor	0.0507
10	UCF	0.0452
11	Colorado	0.0402
12	Brigham Young	0.0345
13	Kansas State	0.0339
14	Texas Tech	0.0203
15	West Virginia	0.0158
16	Utah	0.0122

Fig. 13 Turnovers Steady-State Vector

The full picture is painted, and one can see rankings across all traditional stats. Iowa State (0.1495) finishes in first here, combined with high rankings across all offensive stats, describes an extremely efficient offensive presence on the court. Utah (0.0122), however, finishes last, where they have not broken the top 10 in any category. This is evidence of a weak program on both sides of the court.

Averaged Rankings

No single statistic fully captures an entire program's identity. A team that is a serious scoring threat can struggle at turnovers, illustrating a completely unique identity. To produce a final ranking, we average the 6 individual matrices.

$$A_{final} = (A_{points} + A_{rebounds} + A_{assists} + A_{steals} + A_{blocks} + A_{turnovers})/6$$

As established in the mathematical formulation section, each matrix is column stochastic, so A automatically conforms to the rules of a transition matrix. Then the same power method is applied to the new matrix, producing the final standings steady state vector.

```
============================================
FINAL STANDINGS – MARKOV CHAIN
============================================
  Rk   Team                         pi(i)
  ---- ------------------------- -------
  1    Arizona                    0.1337
  2    Houston                    0.1325
  3    Texas Tech                 0.0986
  4    Kansas                     0.0976
  5    Iowa State                 0.0964
  6    Cincinnati                 0.0676
  7    Brigham Young              0.0570
  8    TCU                        0.0562
  9    Arizona State              0.0481
  10   UCF                        0.0435
  11   Baylor                     0.0404
  12   Colorado                   0.0296
  13   West Virginia              0.0270
  14   Oklahoma State             0.0262
  15   Kansas State               0.0249
  16   Utah                       0.0208
```

Fig. 14 Final Standing Steady-State Vector

Arizona finishes 1st overall (0.1337), with Houston (0.1325) and Texas (0.0986) following behind. Kansas (0.0976), which ranked the highest in the point differential vector, falls to fourth, reflecting how the Markov chain offset Kansas's scoring dominance with marginally weaker results in rebounds, assists, steals, and turnovers (Still 1st in Blocks). Arizona and Houston are consistently among the top of the conference, which shows in the final standings, remarking on how they are well rounded programs by traditional statistical basketball measures.

Compare and Contrast with the 2025-2026 Big 12 Tournament

To test our ranking method and see if it reflects competitive outcomes. The 2025-2026 Big 12 Men's Basketball Tournament had all 16 conference teams participate in high stakes competition.

The top four programs received a double bye, automatically advancing to the quarterfinals, and seeds 5-8 received a first-round bye. The full bracket is below in Figure 15.

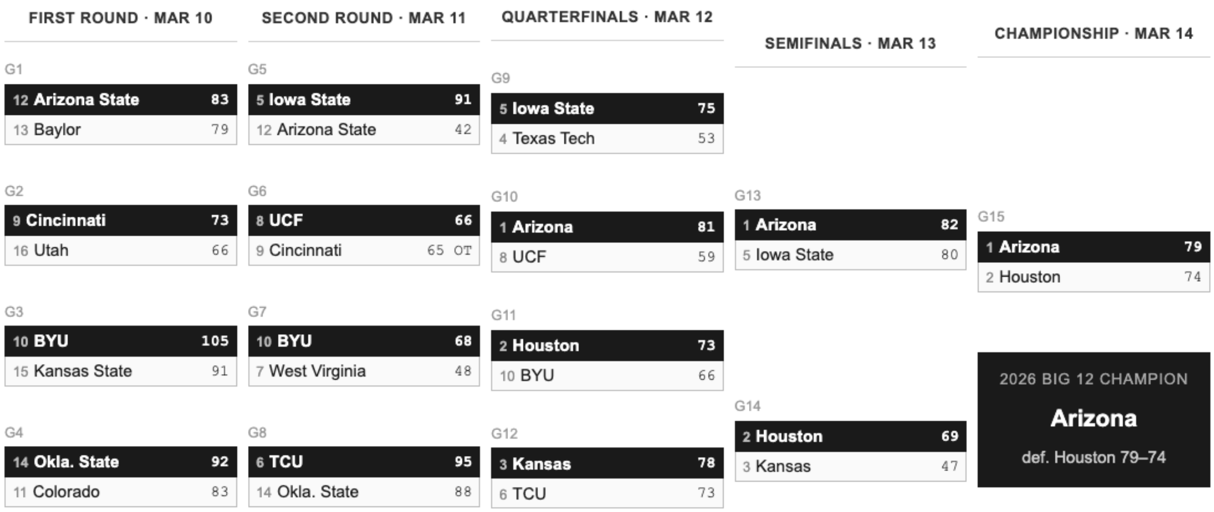

Fig. 15 Big 12 MBB Tournament

With the tournament results, we can compare how our final ranking vector squares up against how well each team actually performed in the tournament.

Final Standings-Markov Chain			Tournament Results
Rank	**Team**	$\pi(i)$	**Seed/Round**
1	Arizona	0.1337	1 – Champion
2	Houston	0.1325	2 – Runner-up
3	Texas Tech	0.0986	4 - Quarterfinals
4	Kansas	0.0976	3 – Semifinals
5	Iowa State	0.0964	5 – Semifinals
6	Cincinnati	0.0676	9 – Second Round
7	Brigham Young	0.0570	10 – Quarterfinals
8	TCU	0.0562	6 – Quarterfinals
9	Arizona State	0.0481	12 – Second Round
10	UCF	0.0435	8 - Quarterfinals
11	Baylor	0.0404	13 – First round
12	Colorado	0.0296	11 - First round
13	West Virginia	0.0270	7 - Second Round
14	Oklahoma State	0.0262	14 - Second Round
15	Kansas State	0.0249	15 - First round
16	Utah	0.0208	16 - First round

Fig. 16 Final Standings Markov Chain vs 2026 Tournament Results

The final ranking has a strong prediction when faced with the tournament. The dark green is an accurate prediction, the light green is 1 away, the yellow is 2 away, the orange is 3 away, and the red is greater than 3. Arizona and Houston, ranked 1st and 2nd respectively by the Markov chain, met in the finals, with Arizona winning 79-74 to win the title. Our methods' top two teams were also the tournament's final two and finished in the same order as our ranking predicted. Kansas (4th in Markov) and Iowa State (5th in Markov) both advanced to the semis before losing to the two finalists, further supporting our rankings.

The least accurate result was Texas Tech, ranked 3rd in our Markov chain, was eliminated in the quarterfinals by Iowa State (53-75). While this data point is not consistent with the final breakdown, looking at the individual vectors can explain this outcome. Texas Tech's program ranked 16th in steals and 14th in turnovers. Whereas, Iowa State ranked 1st in turnovers and 3rd in steals. The Markov chain captured Texas's strengths but couldn't account for this particular stylistic matchup. West Virginia (13rd in Markov) also raises another outlying data point. They are seeded 7 in the tournament's standings and finished in the second round, which means they were eliminated in the 8-12th range.

This poses another problem as an extension of this ranking model. Exploring different weighting ratios when constructing the averaged transition matrix can potentially improve predictions. For example, assigning higher weights to more important statistical measures like point differential can eliminate noise from close games. Weighted versions would potentially yield more accurate tournament results.

Discussions and Conclusions

The rankings using a Markov chain-based model allow for more nuance in the evaluation of team performance in the Big 12 Conference by including several statistical dimensions. This reveals meaningful differences from traditional win-loss standings, including teams which have similar records that are then separated significantly, indicating the win-loss model conceals differences in team strength. By considering the points, rebounds, assists, steals, blocks, and turnovers, the teams that may have high strength in one category but have weaknesses elsewhere may be ranked lower and the more well-rounded teams are ranked higher.

In this model, using the Markov structure, the transition matrices allow indirect comparisons between teams that may not have played equally strong schedules. This allows for schedule strength to be considered in final ranking. Similarly, the steady-state vectors reflect the quality of opponents defeated as opposed to simply a win.

Even with these many advantages, the model does have its limitations. Due to the transition matrix being formed using statistical differentials, outlier games where a team may have an unusual margin of victory could disproportionately adjust the ranking of a team. This could then elevate teams that may have few wins but had a high margin of victory, while not elevating the teams that performed more consistently in closer matches. Similarly, this model does not account for other factors that could impact team performance, such as home-court advantage or player injuries. Finally, including many different statistics and averaging them out provides a better overview of team performance.

The results of the model tested illustrate the importance of including multiple statistics into team evaluation. When comparing each of the steady-state vectors created from the transition matrices, individually, there is much variation in the teams' rankings in each statistic. For example, West Virginia placed very low on turnovers and steals but placed much higher for points. By averaging out all six transition matrices, these differences get moderated and the final ranking becomes more balanced. This suggests that while the individual metrics can capture specific aspects of a team's strength, the combined Markov approach provides a more comprehensive overview. Taken together, these findings demonstrate that team performance in the Big 12 Conference is inherently multi-dimensional and a standard win-loss record cannot provide a comprehensive overview, supporting the use of more sophisticated models for ranking. Future improvements, such as weighting statistical categories or incorporating contextual factors like home-court advantage, could further enhance the model's predictive accuracy and applicability across other sports.

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Appendix

```python
import pandas as pd
import numpy as np

# Load Data
df = pd.read_csv("big12_team_stats_2026.csv")

# Abbreviate Team Names for Printing
TEAM_INFO = [
    ("Arizona",         "UA"),
    ("Arizona State",  "ASU"),
    ("Baylor",         "BAY"),
    ("Brigham Young",  "BYU"),
    ("Cincinnati",     "CIN"),
    ("Colorado",       "CU"),
    ("Houston",        "HOU"),
    ("Iowa State",     "ISU"),
    ("Kansas",         "KU"),
    ("Kansas State",   "KSU"),
    ("Oklahoma State", "OKST"),
    ("TCU",            "TCU"),
    ("Texas Tech",     "TTU"),
```

```
    ("UCF",            "UCF"),
    ("Utah",           "UU"),
    ("West Virginia",  "WVU"),
]

# Normalize name variants in the CSV to match keys
NAME_MAP = {"Texas Christian": "TCU"}

# Get team names and stats
TEAMS  = [t for t, _ in TEAM_INFO]
ABBREV = [a for _, a in TEAM_INFO]
STATS  = ["pts", "reb", "ast", "stl", "blk", "tov"]

team_idx = {t: i for i, t in enumerate(TEAMS)}
n = len(TEAMS)


'''
Construct Transiton matrix for each stat
For each game, compute the stat differential (Team A - Team B) for each stat.
Add the positive differential to the (A, B) entry and 0 to the (B, A) entry.

Normalize the matrix to become column stochastic by dividing each column by its sum.
'''
def build_matrix(stat):
    statMatrix = np.zeros((n, n), dtype=float)

    for url, group in df.groupby("game_url"):
        # ensure grouping is correct (Correct matchup of two teams per game)
        rows = group.reset_index(drop=True)

        if len(rows) != 2:
            continue # skip entries that are incorrectly parsed

        teamA = NAME_MAP.get(rows.loc[0, "team"], rows.loc[0, "team"])
        teamB = NAME_MAP.get(rows.loc[1, "team"], rows.loc[1, "team"])

        # Convert teams to indices
        indA, indB = team_idx.get(teamA, -1), team_idx.get(teamB, -1)
        if indA < 0 or indB < 0:
            continue

        # Compute stat difference
        diff = float(rows.loc[0, stat]) - float(rows.loc[1, stat])
        if stat == "tov":
            diff = -diff # For turnovers, a higher value is worse, so we invert the
difference
        if diff > 0:
```

```
            statMatrix[indA][indB] += diff
        elif diff < 0:
            statMatrix[indB][indA] += abs(diff)

    # Normalize to column stochastic (sum of each column = 1)
    col_sums = statMatrix.sum(axis=0, keepdims=True)
    matrix = np.where(col_sums > 0, statMatrix / col_sums, 0.0)
    return matrix

# Build dictionary of transition matrices for each stat
matrices = {stat: build_matrix(stat) for stat in STATS}

# Label for printing
STAT_LABELS = {
    "pts": "Points",
    "reb": "Rebounds",
    "ast": "Assists",
    "stl": "Steals",
    "blk": "Blocks",
    "tov": "Turnovers",
}

'''
Pretty print the transition matrix for a given stat
Add a header with the stat name and print matrix with a abbreviated team names as
column labels
'''
def print_matrix(stat: str) -> None:
    m = matrices[stat]
    col_w = 6
    row_w = 16

    total_w = row_w + col_w * n
    label = STAT_LABELS[stat]

    # Build border for name of matrix
    print(f"\n{'=' * total_w}")
    print(f"BIG 12 MARKOV TRANSITION MATRIX  --  {label.upper()} DIFFERENTIAL")
    print(f"{'=' * total_w}")

    # Print column headers (abbreviated team names) and add border
    print(f"{'':>{row_w}}" + "".join(f"{a:>{col_w}}" for a in ABBREV))
    print(f"{'-' * total_w}")

    # Print each row with team name and values rounded 3 decimals
    # use '--' for zero entries for better readability
    for i, team in enumerate(TEAMS):
        row_vals = "".join(
```

```
            f"{'--':>{col_w}}" if m[i][j] == 0 else f"{m[i][j]:>{col_w}.3f}"
            for j in range(n)
        )
        print(f"{team:<{row_w}}{row_vals}")

    print("\n" * 2)

# Print all 6 transition matrices for each stat
for stat in STATS:
    print_matrix(stat)

'''
Compute steady state vector for each transition matrix
Find the eigenvector corresponding to eigenvalue 1
If the eigenvector does not satisfy the steady state condition, use power iteration to
find the steady state vector
'''
def steady_state(matrix):
    # Compute eigenvalues and eigenvectors
    eigenvalues, eigenvectors = np.linalg.eig(matrix)

    # Find eigenvalue closest to 1
    idx = np.argmin(np.abs(eigenvalues - 1.0))
    pi = np.real(eigenvectors[:, idx])

    # Make entries non-negative
    pi = np.abs(pi)
    # Normalize to probability vector
    pi = pi / pi.sum()

    # Check if pi is a steady state vector with a small tolerance
    if not np.allclose(matrix @ pi, pi, atol=1e-8):
        # Use power iteration to find steady state vector (since eignvector method
failed to find a valid one)
        # Start with uniform distribution
        pi = np.ones(len(matrix)) / len(matrix)
        for _ in range(100_000):
            # Compute new state and normalize
            pi_new = matrix @ pi
            pi_new /= pi_new.sum()

            # Check for convergence (if the new state is approximately equal to the
old state within a small tolerance)
            if np.allclose(pi_new, pi, atol=1e-8):
                break
            pi = pi_new
    return pi
```

```python
# Pretty print the steady state vector for each stat, ranked from highest to lowest
def print_rankings(title, teams, scores, total_w=60):
    print(f"\n\n{'=' * total_w}")
    print(title)
    print(f"{'=' * total_w}")
    print(f"  {'Rk':<4} {'Team':<25} {'pi(i)':>7}")
    print(f"  {'-'*4} {'-'*25} {'-'*7}")

    # Sort teams by their probability from highest to lowest
    ranked = sorted(zip(teams, scores), key=lambda x: -x[1])

    for rank, (team, prob) in enumerate(ranked, 1):
        print(f"  {rank:<4} {team:<25} {prob:>7.4f}")

    print("\n" * 3)

all_steady = {}
for stat in STATS:
    m  = matrices[stat]
    pi = steady_state(m)
    all_steady[stat] = pi

    print_rankings(f"STEADY STATE VECTOR - {STAT_LABELS[stat].upper()}", TEAMS, pi)

'''
Final Overall Rankings
Find the mean of the 6 transion matrix to create a new composite transiton matrix for
all 6 stats
Compute the steady state vector the composite matrix
'''
M_composite = np.mean(np.stack(list(matrices.values())), axis=0)
pi_composite = steady_state(M_composite)

print_rankings("FINAL STANDINGS - MARKOV CHAIN", TEAMS, pi_composite)
```